Technical Report: Deep Learning for an 11-year-old

Topic: Weights, Bias, and Activation Functions

Assignment: A03 ITAI 2376 Deep Learning for an 11-year-old

Course: ITAI 2376 Deep Learning Artificial Intel

Institution: Houston City College

Date: February 11, 2026

Team Name: Team 1

Team Members: DeMarcus Crump, Chloe Tu, Matthew Choo, Franck eric Kolontchang toheu

THE NEURAL NETWORK SQUAD - A Deep Learning Adventure

Website Link (Interactive Web Comic): https://neural-network-squad.netlify.app/

1. Introduction

Have you ever wondered how a computer can look at a picture and say, "That's a cat!" or how your video game knows when you scored a goal? It all happens because of something called a **neural network**.

Think of a neural network like a team of tiny helpers inside a computer. Each helper is called a **neuron**. Just like your brain has neurons that help you think, computers have artificial neurons

that help them learn and make decisions.

In this report, we will explain three important parts that help every neuron do its job:

- **Weights** – They decide how important something is
- **Bias** – It gives the neuron a head start
- **Activation Functions** – They are the gatekeepers who make the final decision

Our goal is to explain these ideas so that any 11-year-old can understand them. To make it fun, we created an interactive comic book that shows how these parts work together!

2. Activities Performed

Our team worked together to make this topic fun and easy to understand. Here is how we split up the work:

- **DeMarcus** created the interactive web comic storybook
- **Franck** researched and explained Weights
- **Chloe** researched and explained Bias
- **Matthew** researched and explained Activation Functions

We each studied our topic carefully, then worked together to turn hard math words into simple ideas that kids can understand.

3. Research Findings and Analysis

3.1 Background

The main goal of this assignment was to pick a hard topic in AI and teach it to a younger student. We chose the artificial neuron because it is the building block of everything in Deep Learning.

3.2 Key Features

Our research focused on three big words: Weight, Bias, and Activation Function. These sound hard, but they are actually simple tools that the computer uses. We turned these math terms into easy ideas by comparing them to things kids see every day, like video games and volume knobs.

4. The Problem: Why Do Computers Get Confused?

Imagine you are playing the telephone game with a million friends. By the time the message gets

to the last person, it is totally wrong! That is what happens inside a computer when millions of signals flow through at once. Everything gets messy.

A computer might look at a picture and think, "Is that a cat... or a toaster?!" It cannot tell because all the information is jumbled up.

That is why we need Weights, Bias, and Activation Functions. When they work together, they help the computer understand the message clearly.

5. The Computational Flow

A neuron works like a factory line. It follows a set of steps to get an answer:

1. Input: The neuron catches data, like catching a ball.

2. Weighted Sum: It multiplies the data by a number called the Weight.

3. Bias Addition: It adds an extra number called the Bias to the total.

4. Activation: The Activation Function looks at the number and makes a final decision.

5. Output: The neuron sends its answer to the next neuron in the line.

The Formula: OUTPUT = Activation (Weight X Input + Bias)

6. Real-World Applications

To make this real for students, we created a cartoon storybook. This helps kids see how math actually works in real life. It shows that these numbers are not just on a page. They help computers do things like:

- Recognize faces in photos
- Understand your voice when you talk to Siri or Alexa
- Play video games against you
- Help self-driving cars see the road

7. Core Components

This section breaks down the three parts of the neuron we researched. We turned the math into plain English.

7.1. Weights (The Importance Factor) - Researched by Franck

What is a Weight? A weight is a number that multiplies an input signal to make it stronger or weaker.

Simple Explanation: Think of weights like the volume knobs on a radio or mixing board. Turn it up, and the sound gets louder. Turn it down, and it gets quieter. Weights do the same thing with information!

Here is how weights work:

- **High Weight (like 2.0)** = "This is SUPER important!" The signal gets bigger.
- **Low Weight (like 0.5)** = "This is not very important." The signal gets smaller.
- **Negative Weight (like -1)** = "Flip it around!" The signal changes direction.

Example: If you have an input of 5 and a weight of 2, the result is . The signal got stronger!

7.2. Bias (The Activation Offset) - Researched by Chloe

What is a Bias? A bias is a number that gets added after the weights do their work. It shifts the starting point.

Simple Explanation: Think of bias like getting bonus points before a test even starts. Even if you start with zero, the bias gives you extra points to help you out! You can also think of it like a head start in a race.

Here is how bias helps:

- **Positive Bias (like +3)** = You get a head start! The number goes up.
- **Negative Bias (like -2)** = You start a little behind. The number goes down.

Why we need it: Without bias, the computer might get stuck at the starting line. Bias makes sure the neuron can work even when the input is zero or very small.

Example: If the weighted signal is **10** and the bias is **+3**, the result is **10 + 3 =13**. It got a boost!

7.3. Activation Functions (The Decision Maker) - Researched by Matthew

What is an Activation Function? An activation function takes the signal (after weights and

bias) and transforms it. It decides what actually gets through and makes the final call.

Simple Explanation: Think of activation functions like bouncers at a club or a light switch. They decide who gets in and how!

Without activation functions, computers could only learn straight lines. Activation functions add curves, which help computers learn tricky patterns like recognizing faces or understanding words.

There are three main types:

- **Sigmoid**
  - Squishes everything between 0 and 1.
  - Perfect for yes-or-no questions like "Is this a cat?"
- **Tanh (Hyperbolic Tangent)**
  - Like Sigmoid, but keeps things balanced between -1 and +1.
  - Great for things that can be negative, neutral, or positive.
- **ReLU (Rectified Linear Unit)**
  - The most popular one used today!
  - Simple rule: Positive numbers? Let them through! Negative numbers? BLOCKED (turns them to zero)!
  - This makes the computer faster and more efficient.

8. Conclusion: What was Learned

We learned a lot about Weights, Bias, and Activation Functions. Here is what we discovered:

1. Weights decide how important each piece of information is (like volume knobs).

2. Bias gives the neuron a head start or adjustment (like bonus points).

3. Activation Functions are the gatekeepers that make the final decision (like bouncers or light switches).

The most important lesson is that these three parts cannot work alone. Just like a sports team, they need to work **TOGETHER** in the right order to win!

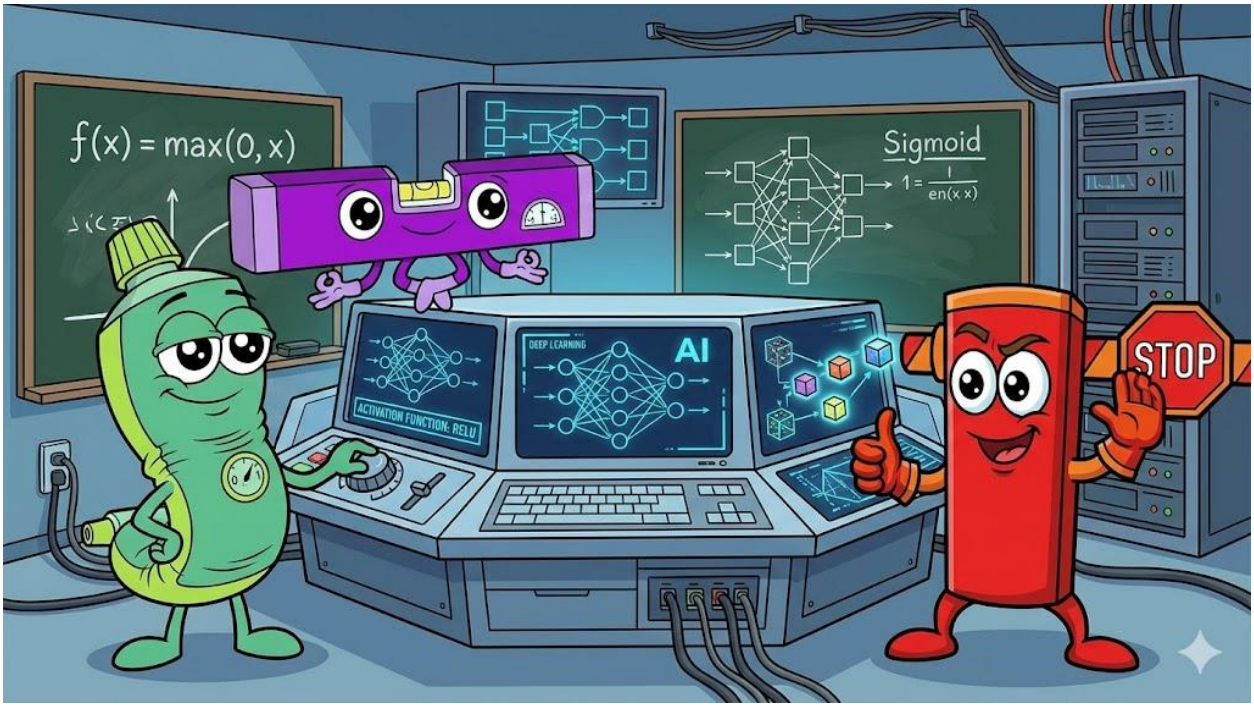
One neuron can solve simple problems. But Deep Learning uses millions of these neurons stacked in layers. By adjusting Weights and Biases, the computer can learn to do amazing things like recognize faces, understand language, and even drive cars!

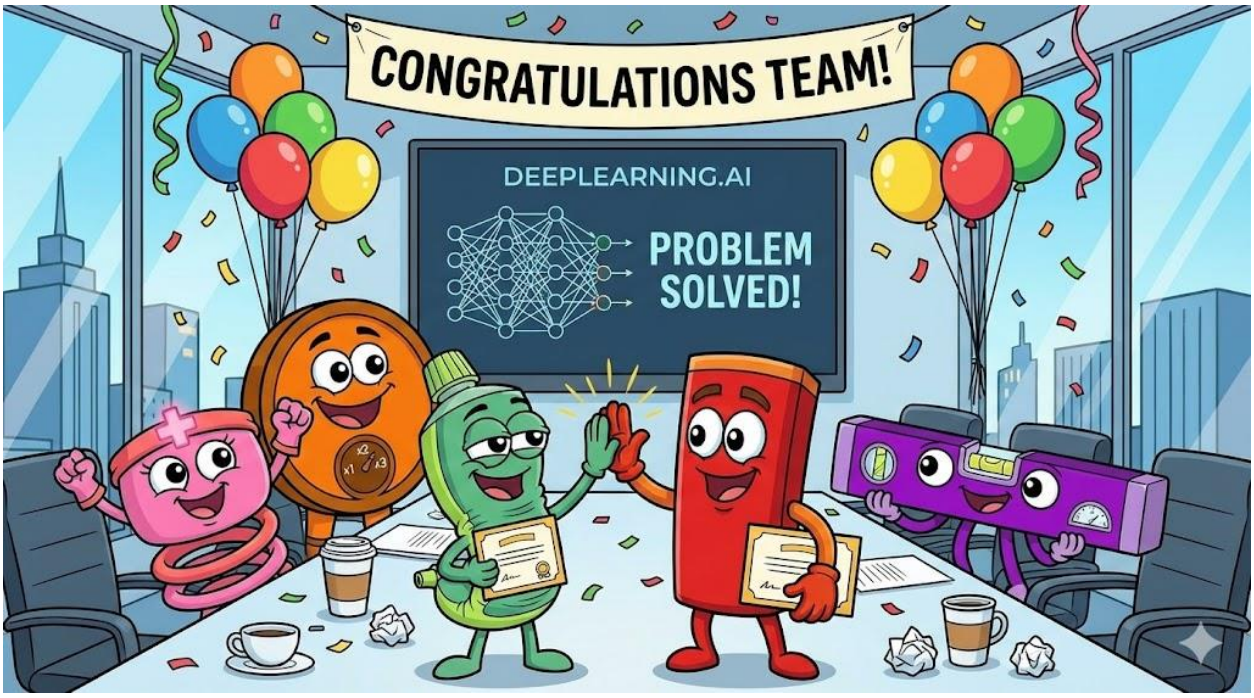
9. Media Gallery

To help explain these ideas to an 11-year-old audience, our team created a visual learning hub on the web.

The Neural Network Squad Website (Interactive Web Comic)

Description: This website hosts the creative materials we developed for this project. It serves as a colorful, interactive guide where students can see the concepts of Weights, Bias, and Activation Functions in action.





The app includes:

- 10 pages of colorful comic panels
- Cartoon characters that represent each neural network component
- Interactive sliders so you can try weights and bias yourself
- Definition boxes that explain each term in simple words
- A glossary with all the important words

10. References

- Baheti, Pragati. "12 Types of Neural Networks Activation Functions: How to Choose?" Www.v7labs.com, 8 Mar. 2022, www.v7labs.com/blog/neural-networks-activation-functions. Accessed 29 Jan. 2026.
- Luna, Javier Canales. "A Beginner's Guide to the Rectified Linear Unit (ReLU)." Datacamp.com, DataCamp, 28 Jan. 2025, www.datacamp.com/blog/rectified-linear-unit-relu. Accessed 29 Jan. 2026.
- Luca, Gabriele De. "Baeldung." Baeldung on Computer Science, 20 July 2020, www.baeldung.com/cs/neural-networks-bias. Accessed 29 Jan. 2026.
- GeeksforGeeks. "Effect of Bias in Neural Network." GeeksforGeeks, 25 Sept. 2018,

www.geeksforgeeks.org/deep-learning/effect-of-bias-in-neural-network/. Accessed 2 Feb. 2026.

- Turing. "Importance of Neural Network Bias and How to Add It." Www.turing.com, www.turing.com/kb/necessity-of-bias-in-neural-networks. Accessed 29 Jan. 2026.
- Ali, Moez. "Introduction to Activation Functions in Neural Networks." Datacamp.com, DataCamp, 9 Nov. 2023, www.datacamp.com/tutorial/introduction-to-activation-functions-in-neural-networks. Accessed 29 Jan. 2026.
- "Neural Networks: Activation Functions." Google for Developers, 2024, https://developers.google.com/machine-learning/crash-course/neural-networks/activation-functions. Accessed 29 Jan. 2026.
- GeeksforGeeks. "Weights and Bias in Neural Networks." GeeksforGeeks, 11 Oct. 2023, www.geeksforgeeks.org/deep-learning/the-role-of-weights-and-bias-in-neural-networks/. Accessed 29 Jan. 2026.